

Supplementary Materials: Blurb-Refined Inference from Crowdsourced Book Reviews using Hierarchical Genre Mining with Dual-Path Graph Convolutions

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Dataset: Genre Distribution

Our dataset is a pioneering effort in two-level hierarchical genre classification, comprising 9375 books with associated blurb information and 90499 user reviews. At level-1, the books are broadly categorized into two primary categories: *fiction* and *nonfiction*, comprising 5612 and 3763 books, respectively. Each book within these main categories is further annotated with one or more genres at level-2, which includes 28 sub-genres for fiction and 29 for nonfiction. Notably, the level-2 annotations follow a multi-label format. The complete list of level-2 genres is presented in Table 1.

Building upon the earlier Book-Crossing dataset (Ziegler et al. 2005), which often overlooks the inherent hierarchical and multi-label structure of genre taxonomies, we constructed a new dataset by crawling Goodreads (www.goodreads.com). We collected both the blurb and user reviews for each book. While the blurbs were largely consistent and aligned well with the central themes of the books (with only a few instances of missing blurbs), user reviews tended to be noisy and occasionally inconsistent. Furthermore, to annotate the dataset, we began with an initial set of flat genre categories collected from Goodreads, which often numbered in the thousands. To ensure high-quality annotation and maintain the integrity of the dataset, we enlisted five Master’s or PhD-level students with foundational knowledge in linguistics. Under the supervision of a linguistics expert, these annotators were rigorously trained to label the data following strict guidelines. The heat maps in Fig. 1 (a) and (b) illustrate the label co-occurrence and sparsity patterns among fiction and nonfiction genres, respectively, based on the final annotated dataset.

Following annotation, while developing the genre prediction tool, HiGeMine, our proposed framework for hierarchical genre classification, we observed a severe data imbalance issue in the training set. To address this, we performed data augmentation on both the blurbs and user reviews to generate artificial samples, thereby improving training effectiveness and mitigating overfitting. Table 1 presents the distribution of the augmented data.

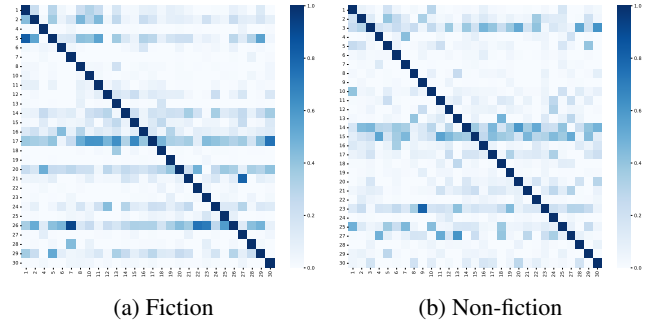


Figure 1: Label co-occurrence heatmap of our dataset, where each cell indicates the probability of the column-label occurring under the row-label, indicating the clear picture of the relationship among the labels in the dataset.

Table 1: Distribution of genre categories in the dataset

Genre ID	Genre name	Fiction		Non-fiction	
		Org.	Aug.	Org.	Aug.
1	Animals & Wildlife & Pets	542	648	183	598
2	Arts & Photography	1026	1180	483	674
3	Biographies & Memoir	-	-	1145	348
4	Business & Money	40	699	235	526
5	Children’s Book	1525	1325	237	736
6	Comics & Graphic	328	316	92	516
7	Computers & Technology	25	420	75	546
8	Cookbooks & Food & Wine	69	763	167	618
9	Crafts & Hobbies & Home	-	-	43	566
10	Environment & Plant	84	537	256	548
11	Family & Parenting & Relationships	200	535	244	398
12	Fashion & Lifestyle	609	341	163	580
13	Health & Fitness & Dieting	30	476	269	576
14	History	1354	1183	1398	610
15	Humanities	441	433	1349	438
16	Humor & Entertainment	881	667	332	332
17	Literature	2183	1035	532	306
18	Medical	62	664	152	522
19	Meta Text	33	778	81	604
20	Mystery & Thriller & Suspense & Horror	1752	917	358	798
21	Mythology & Religion & Spirituality	707	884	654	394
22	Press & Media	130	471	145	612
23	Reference & Language	84	548	804	586
24	Romance	1171	677	70	590
25	Science & Math	349	252	548	428
26	Science Fiction & Fantasy	2257	835	-	-
27	Self-help & Motivation	66	700	590	414
28	Sports & Outdoors	153	483	127	606
29	Teen & Young Adult	1435	454	165	496
30	Travel	78	689	334	456
Genre Count		5612	12149	3763	11739

Org.: # Original, Aug.: # Augmented

Table 2: Brief details of hyperparameters

Attributes	Values
Optimizer	AdamW (Loshchilov et al. 2019)
Learning Rate	Level-1: 10^{-4} Level-2: 10^{-3}
λ_1 & λ_2	[0, 1] & [0, 1]
LM Encoder Feature	768
GloVe Encoder Feature	Dimension: 300
Learnable Embedding Layer	300
GCN ($\mathcal{B}$ - $\mathcal{T}$, $\mathcal{P}$ - $\mathcal{T}$, and label co-occurrence network)	No. of layers: 2 Dimension: 128
Level-1 Dense Network	No. of layers: 2 Dimension: 128 & 2
Level-2 Dense Network	No. of layers: 1 Dimension: 128

Hyperparameters Configuration

Table 2 presents the configuration of the hyperparameters used in our experiments. We utilized pretrained language model (LM) encoders, which were fine-tuned for a few epochs depending on the task, binary or multi-label classification, aligned with the hierarchical structure of HiGeMine. The fine-tuned LM encoders are subsequently used for training the remaining components of the framework. A consistent dropout rate of 0.1 was applied across all models. To prevent overfitting, we employed early stopping with a patience of 5 epochs based on validation performance, and set a maximum training limit of 200 epochs.

Empirical Study: Language Models (LMs)

HiGeMine primarily relies on an LM encoder to generate high-dimensional embeddings of book blurbs and user reviews. In the review filtering stage, we use the pretrained LM (without any fine-tuning) to extract filtered reviews that are more aligned with the blurb text and less noisy. In the second phase, classifier training, we explicitly used a fine-tuned LM, whose parameters are tuned prior to the main framework training. Specifically, we adopt the BERT-base language model (Devlin et al. 2019) in both stages.

BERT is a foundational LM, based on the Transformer encoder, and comprises nearly 110 million parameters. It is pre-trained on large-scale internet corpora with dual objectives, masked language modeling and next sentence prediction. Its bidirectional encoding capability allows it to effectively capture rich contextual dependencies from the input text.

To evaluate the effectiveness of BERT-based encoders and to justify our choice of using BERT in our case, we include five additional LM-based encoders, detailed as follows:

- *ALBERT* (Lan et al. 2020): A lightweight variant of BERT designed to reduce memory consumption and improve training efficiency. It achieves this by factorizing embeddings and sharing parameters across layers. ALBERT contains 18 $\times$ fewer parameters and trains 1.7 $\times$ faster than BERT.
- *DistilBERT* (Sanh et al. 2019): A compressed version of BERT that employs knowledge distillation during pre-

Table 3: Performance of HiGeMine with LM encoders

Models	Level-1		Level-2					
			Fiction			Non-Fiction		
	$\mathcal{F} \uparrow$	$\mathcal{A}c \uparrow$	$\mathcal{F}_\mu \uparrow$	$\mathcal{F}_m \uparrow$	$\mathcal{H}\mathcal{L} \downarrow$	$\mathcal{F}_\mu \uparrow$	$\mathcal{F}_m \uparrow$	$\mathcal{H}\mathcal{L} \downarrow$
ALBERT	87.41	90.42	68.82	59.97	0.08303	74.48	71.60	0.06082
XLNet	88.92	91.11	75.01	64.50	0.06326	69.42	65.16	0.08061
RoBERTa	92.43	93.93	77.95	69.08	0.05414	76.99	73.22	0.05425
ModernBERT	94.36	95.42	65.20	52.56	0.10426	60.53	54.75	0.11945
DistilBERT	95.24	96.17	79.58	74.88	0.04857	80.45	76.05	0.04454
BERT	96.91	97.50	81.89	77.24	0.0436	81.86	78.57	0.0401

training. It retains 97% of BERT’s language understanding capabilities while using 40% fewer parameters and achieving 60% faster inference.

- *ModernBERT* (Warner et al. 2024): An optimized BERT variant that introduces a revised tokenization scheme and improved training procedures to enhance adaptability, computational efficiency, and model robustness.
- *RoBERTa* (Zhuang et al. 2021): A robustly optimized variant of BERT that removes the next-sentence prediction objective, allowing for training over larger batches and longer sequences on more extensive datasets.
- *XLNet* (Yang et al. 2019): A generalized autoregressive pretraining method that combines the strengths of autoregressive and autoencoding objectives through permutation-based training. While powerful, XLNet is more challenging to fine-tune than BERT due to its complex training strategy.

Although BERT is more computationally demanding than some of the optimized baselines, we adopt it in HiGeMine due to its strong trade-off between accuracy, generalization, and ease of fine-tuning, as demonstrated in our experimental results. Its bidirectional encoding and ability to capture rich contextual semantics make it particularly effective for book genre classification.

For a fair comparison, all models were trained using the same configuration as BERT-base. Table 3 reports the performance of the HiGeMine framework with BERT-base and the five alternative LM encoders on the metrics F1-score $\mathcal{F}$ and Accuracy $\mathcal{A}c$ at Level-1, and macro and micro-averaged F1-score and Hamming Loss $\mathcal{H}\mathcal{L}$ at Level-2.

BERT consistently outperforms the other models across all evaluation metrics. Notably, using ALBERT (Lan et al. 2020), the weakest performer, resulted in a substantial performance drop, with $\mathcal{F}$ decreasing by 10.87% and $\mathcal{A}c$ by 7.83% at Level-1. This decline is likely attributable to ALBERT’s compressed architecture, which limits its ability to capture nuanced semantic representations. Although ModernBERT (Warner et al. 2024) outperformed ALBERT at Level-1 (gains of 7.95% in $\mathcal{F}$ and 5.53% in $\mathcal{A}c$), ALBERT showed stronger performance at Level-2, indicating task-dependent behavior. DistilBERT (Sanh et al. 2019), the second-best model, trailed BERT closely at Level-2 across most metrics, yet consistently lagged overall. These observations affirm BERT as the most effective encoder for our framework.

Table 4: Performance of HiGeMine with baselines

Category	Models	Fiction (F)				Non-Fiction (NF)			
		$\mathcal{F}_L$	$\mathcal{B}A_u$	$\mathcal{F}_m$	$\mathcal{B}A_m$	$\mathcal{F}_L$	$\mathcal{B}A_u$	$\mathcal{F}_m$	$\mathcal{B}A_m$
SOTA-B	Scotfield et al.	11.99	53.19	5.97	51.57	7.80	52.03	10.12	52.77
	N.-Flores et al.	29.85	59.09	14.18	54.56	27.86	58.28	24.30	57.69
	Ullah et al.	39.62	63.28	18.35	56.31	39.84	63.18	32.55	61.30
	Improvement (%)	106.69	46.65	320.93	64.25	105.47	48.78	141.38	52.09
SOTA-R	Ullah et al.	43.65	65.03	21.40	57.33	47.65	67.02	39.69	64.54
	Scotfield et al.	50.89	67.29	41.71	64.04	40.29	62.70	33.64	60.68
	N.-Flores et al.	53.05	69.66	35.25	63.99	60.14	75.05	56.65	74.19
	Improvement (%)	54.36	33.22	85.18	44.43	36.12	25.25	38.69	25.66
HTC-based	HiAGM	64.13	80.47	47.95	73.31	64.04	80.13	57.82	78.63
	HiTIN	73.84	86.04	62.90	80.88	74.40	86.88	68.90	84.47
	HILL	75.90	83.70	76.23	84.79	76.19	83.53	76.73	85.06
	Improvement (%)	7.89	7.86	1.32	9.08	7.44	8.20	2.40	9.64
LM-based	ALBERT	58.00	78.60	47.25	72.95	63.75	84.86	61.49	81.80
	BERT	65.23	82.02	57.43	76.92	72.31	88.16	67.53	83.82
	DistilBERT	67.39	83.35	55.78	77.45	73.25	87.00	68.77	84.23
	ModernBERT	59.94	79.02	50.94	73.28	69.51	86.62	65.11	82.98
	RoBERTa	65.88	80.89	51.49	74.76	72.07	87.97	66.46	83.77
	XLNET	63.46	81.24	52.26	74.40	69.68	86.05	66.11	83.04
	Improvement (%)	21.52	11.34	38.47	19.42	11.75	6.62	14.25	10.69
Open-source LLM	Qwen3-4B	74.71	82.23	54.42	72.86	74.19	83.30	66.60	78.78
	Gemma2-2B	75.45	83.70	58.41	76.22	75.24	83.48	70.12	80.51
	Mistral-7B	81.20	87.90	69.51	81.43	78.79	86.68	73.27	83.86
	Improvement (%)	0.85	5.57	11.12	13.58	3.90	8.44	7.23	11.17
Close-source LLM	Deepseek-V3	62.97	82.97	57.25	83.16	64.75	80.03	41.70	70.47
	Gemini-2.5-flash	59.11	84.41	51.79	82.85	62.57	81.73	39.03	72.26
	GPT-4.1-mini	64.08	81.89	58.70	82.36	65.01	80.38	41.10	70.15
	Improvement (%)	25.96	9.94	31.58	11.21	25.91	15.01	88.44	29.03
HiGeMine		81.89	92.80	77.24	92.49	81.86	94.00	78.57	93.23

Second-best method in each category is underlined

Baseline Evaluation

The main paper presents the overall results for the baselines; here, we include more fine-grained results to offer deeper insights into model behavior and performance across different baselines. Table 4 summarizes these extensive experiments on metrics including macro and micro-averaged, F1-score and Balance Accuracy $\mathcal{B}A$, improvements highlighted with gray color in the last row of each category.

(i) *Performance comparison with previous genre classifier methods:* Book genre classification remains relatively underexplored. Early studies have typically relied on either blurbs or reviews, focusing on languages such as Spanish (Flores et al. 2023), Portuguese (Scotfield et al. 2022), and Bangla (Ullah et al. 2023). These methods often address the genre prediction without accounting for hierarchical structures; therefore, lacks the natural taxonomies within the literary. The first two segments of Table 4, SOTA-B (blurbs only) and SOTA-R (reviews only), present results corresponding to these settings. Overall, methods using only reviews consistently outperform their blurb-only counterparts. While (Ullah et al. 2023) performs best among blurb-based methods, it still lags behind review-based ones. In contrast, our proposed model, HiGeMine, which fuses both modalities (reviews and blurbs), exceeds both individual settings by a clear margin, as shown in the last row of the first two segments.

(ii) *Performance comparison with hierarchical text classification methods (HTC):* Owing to the novel hierarchical design of HiGeMine, it is essential to compare it against existing hierarchical text classification (HTC) methods. We therefore evaluate three representative HTC models, HiAGM (Zhou et al. 2020), HiTIN (Zhu et al. 2023), and HILL (Zhu et al. 2024), with results reported in the HTC-based section of Table 4. Among these, HILL shows the strongest overall performance; however, HiTIN achieves a higher micro-averaged balanced accuracy, likely due to its ability to recover more true positives per class (particularly for under-represented classes), albeit at the cost of lower pre-

cision, resulting in poorer F1 values. In contrast, HiGeMine consistently surpasses all three methods, achieving a superior balance in the precision-recall trade-off.

(iii) *Performance comparison with LM-based methods:* Foundational LMs have recently achieved remarkable success across various domains. Accordingly, we compare HiGeMine against six prominent LM-based baselines: ALBERT (Lan et al. 2020), BERT (Devlin et al. 2019), DistilBERT (Sanh et al. 2019), ModernBERT (Warner et al. 2024), RoBERTa (Zhuang et al. 2021), and XLNet (Yang et al. 2019). To ensure a fair comparison, we maintain the hierarchical architecture for all models, using an LM followed by a two-layer MLP classifier at both Level-1 and Level-2. To combine blurb and review embeddings, we evaluate multiple feature fusion strategies (addition, concatenation, self-attention, and cross-attention), and report only the best-performing fusion setting for each LM in Table 4. Among the baselines, ALBERT yields the weakest performance, whereas DistilBERT performs best overall, with BERT closely competitive. Notably, HiGeMine consistently outperforms all LM-based classifiers across all evaluation metrics, highlighting the benefit of incorporating graph-based learning and modeling both word-text context and label correlations.

(iv) *Performance comparison with Open-source Large Language Models (LLMs):* LLMs are powerful tools, yet they often struggle with tasks requiring complex reasoning. To evaluate HiGeMine in this context, we compare it with open-source LLMs fine-tuned on our dataset in the same manner as the LM-based baselines. Specifically, we include three recent models with increasing parameter sizes: Gemma2-2B (Gemma Team 2024), Qwen3-4B (Qwen Team 2025), and Mistral-7B (Jiang et al. 2023). Overall, HiGeMine consistently outperforms Mistral-7B, the strongest among the open-source models, demonstrating its ability to effectively balance performance and computational efficiency.

(v) *Performance comparison with Close-source methods:* Finally, we evaluate HiGeMine against three closed-source LLMs, Deepseek-V3 (DeepSeek-AI et al. 2024), Gemini-2.5-flash (Comanici et al. 2025), and GPT-4.1-mini (OpenAI et al. 2024), accessed via API. HiGeMine consistently outperforms all closed-source baselines across all evaluation metrics in both genre categories. Overall, it offers robust and reliable performance while substantially reducing the computational cost associated with LLMs.

Qualitative Analysis: Review Filtering

Fig. 2 presents a qualitative analysis demonstrating the efficacy of review filtering for genre classification. The green highlighted reviews represent those selected by our filtering mechanism based on their relevance to informational content. These selected reviews contain substantial genre-indicative information about the book. For instance, Fig. 2 provides genre through phrases, *Review 2*: “a boy living in the prehistoric era who wants to be a cave painter”, “middle grade level”, “historical detail and adventure”, and characterizes it as a short, fun, historical fiction read. Such phrases

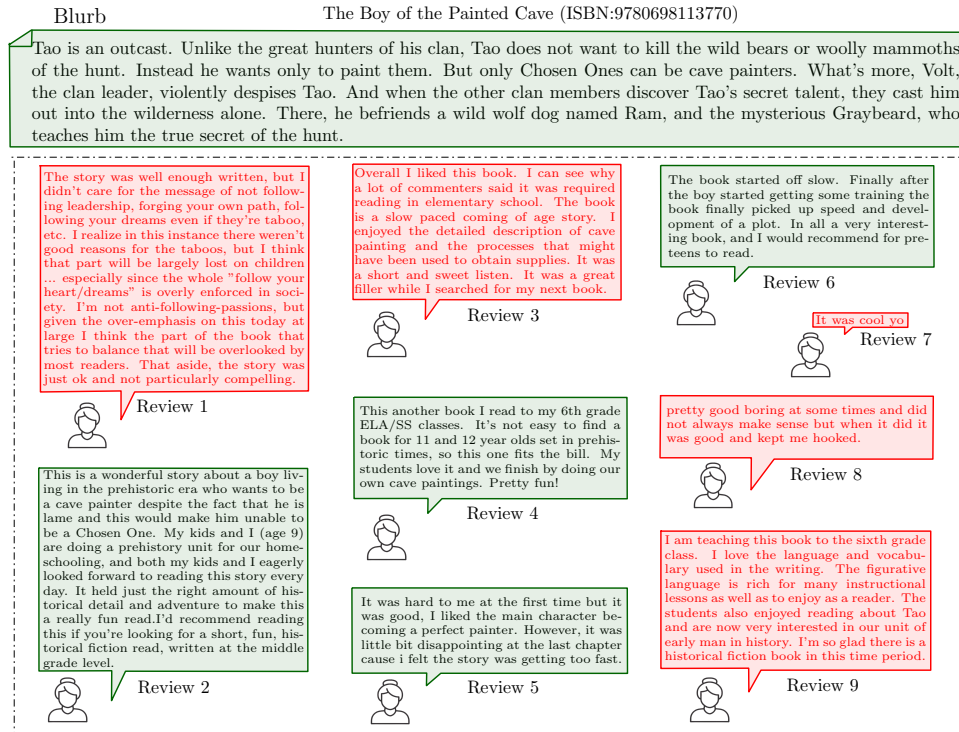


Figure 2: Qualitative analysis for review filtering for a sample blurb. Filtered-in and filtered-out reviews are highlighted in green, and red color, respectively.

offer information for genre classification, specifically indicating alignment with *fiction*, *children's book*, and *history* categories.

In contrast, reviews that were not selected exhibit significant information sparsity, exemplified by the review, *Review 7*: “It was cool yo”, which does not provide genre-relevant content. Further, the filtering process systematically excludes reviews that contain potentially misleading information, including personal opinions, *Review 3*: “Overall I liked this book” or offering generic reviews without genre-specific content, *Review 8*: “pretty good boring at some times”. Such reviews focus primarily on the reader experience rather than the characteristics of the content.

This review filtering approach ensures that HiGeMine focuses on high-quality, noise-free, genre-relevant data, thereby boosting accuracy and performance.

Modality Specific Qualitative Analysis

In this section, we provide the qualitative analysis of HiGeMine, exploiting the impact of modality.

Fig. 3 shows the qualitative analysis of HiGeMine classification ability through three sample scenarios (*a*, *b*, *c*) involving book blurbs and reviews. The analysis compares predictions across four model configurations with varying modality weightings (λ_1 , λ_2): HiGeMine (1, 1), HiGeMine (0, 0), HiGeMine (0.5, 0.5), and standard HiGeMine.

- **Sample (a)**, *Information-spare blurb with comprehensive reviews*: HiGeMine (1, 1) failed to correctly classify gen-

res due to over-reliance on the uninformative blurb, while other configurations succeeded by effectively leveraging review content. This highlights the model's adaptability when the blurb is inadequate.

- **Sample (b)**, *Informative blurb with limited reviews*: HiGeMine (0, 0) was completely misclassified as it only used limited reviews, and HiGeMine (1, 1) achieved partial accuracy. Other configurations demonstrated full accuracy through the integration of both modalities.
- **Sample (c)**, *Sufficient content in the blurb and reviews*: Standard HiGeMine achieved perfect genre prediction by complementary from both blurb and reviews. At the same time, other configurations showed reduced performance due to suboptimal modality weighting.

This demonstrates standard HiGeMine's superior cross-modal integration capabilities compared to its variants, particularly in scenarios requiring dynamic adaptation to variable information quality across blurb and reviews.

Misprediction Analysis

We posit that genre co-occurrence patterns play a major role in mispredictions. In this section, we analyze misclassifications arising either from indistinguishable phrasing within blurbs or reviews, or from frequent genre co-occurrence as follows:

Fig. 4 presents a qualitative analysis of HiGeMine's classification errors caused by misleading genre information

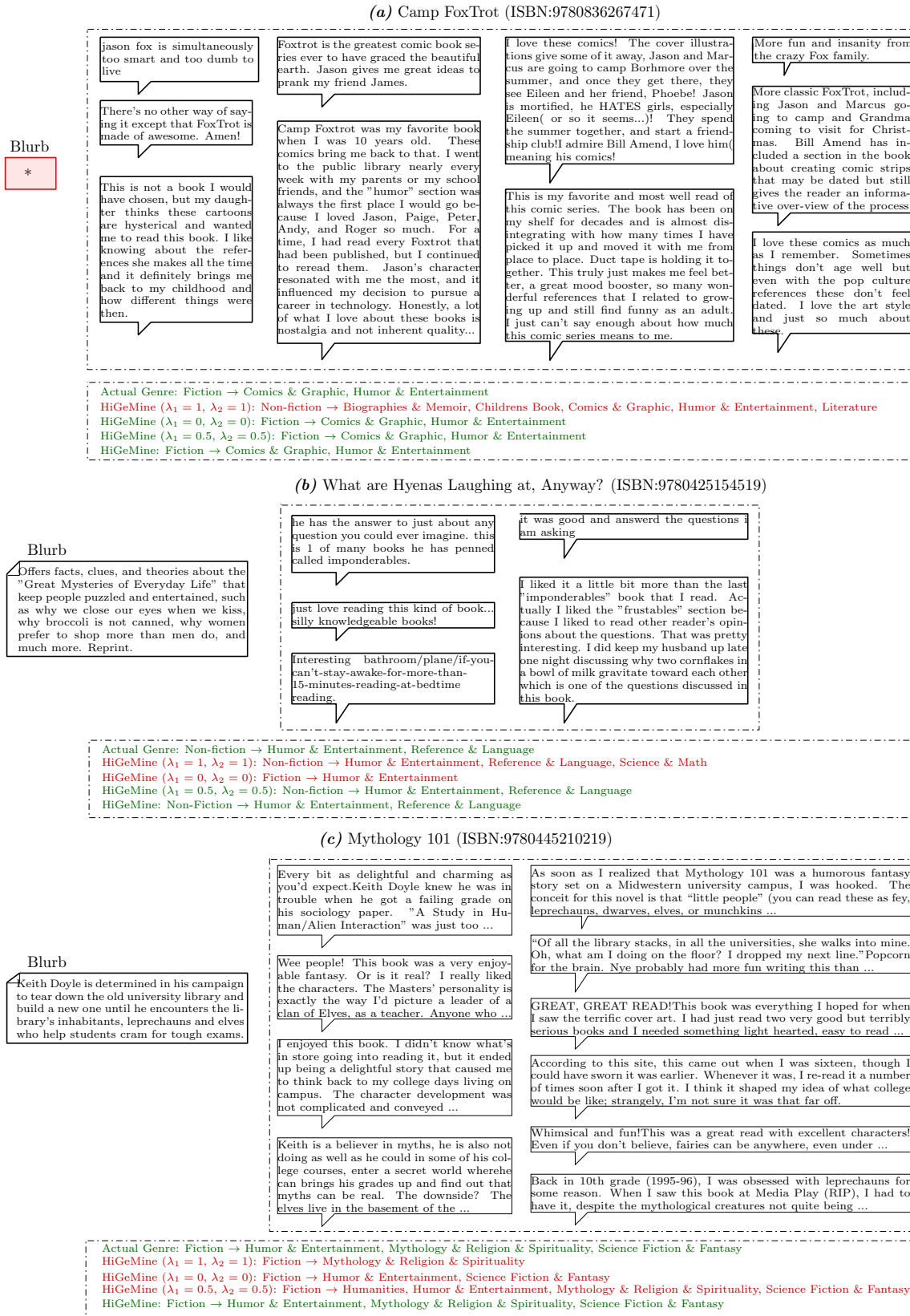


Figure 3: Qualitative analysis of modality ablation: **(a)** Scarce blurbs lead to incorrect predictions, while reviews alone or combined with blurbs yield correct genres; **(b)** Informative blurbs and scarce reviews may still mislead individually, but their combination enables correct classification; **(c)** Even equal contributions from both modalities may fail, whereas HiGeMine adaptively balances them to predict accurately. Note that a few reviews are truncated due to over length.

Table 5: Fiction genre misprediction analysis

(a) Sample Count																																	
ID ↓→	1	2	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
1	399	176	0	331	8	0	15	23	34	0	5	21	9	52	152	2	2	94	17	0	0	8	4	139	3	0	144	0					
2	176	748	1	419	11	4	29	20	69	17	7	49	12	149	261	3	3	125	45	4	15	63	11	381	7	10	172	5					
4	0	1	28	0	1	1	0	0	0	0	0	6	5	2	11	0	0	13	2	1	0	0	0	4	4	0	0	1					
5	331	419	0	1119	31	3	26	30	85	0	10	150	20	199	458	6	6	269	109	3	7	44	2	465	8	33	581	3					
6	8	11	1	31	222	1	1	0	0	0	0	23	16	92	8	0	0	53	0	3	1	26	3	103	0	1	20	0					
7	0	4	1	3	1	18	0	0	0	0	0	1	1	2	6	0	0	5	0	0	1	1	1	16	0	6	1	1					
8	15	29	0	26	1	0	55	2	4	5	4	2	3	16	27	0	0	11	2	1	0	9	2	11	0	0	3	3					
10	23	20	0	30	0	0	2	70	5	1	0	14	11	5	43	0	0	15	8	0	0	4	5	21	0	4	8	3					
11	34	69	0	85	0	0	4	5	150	14	2	37	0	20	93	4	0	5	4	0	0	25	0	9	3	0	29	1					
12	0	17	0	0	0	0	5	1	14	434	3	121	71	45	205	5	2	125	15	0	1	181	0	100	9	0	12	8					
13	5	7	0	10	0	0	4	0	2	3	23	1	6	3	13	8	0	4	1	0	0	0	5	0	5	0	8	0					
14	21	49	6	150	23	1	2	14	37	121	1	928	66	37	303	11	5	252	159	17	1	263	31	261	13	6	237	20					
15	9	12	5	20	16	1	3	11	0	71	6	66	324	37	210	12	2	70	45	0	4	39	1	106	14	2	32	9					
16	52	149	2	199	92	2	16	5	20	45	3	37	37	624	251	3	7	116	10	10	6	59	16	204	3	7	114	6					
17	152	261	11	458	8	6	27	43	93	205	13	303	210	251	156	25	9	395	176	10	29	333	89	502	22	16	443	40					
18	2	3	0	6	0	0	0	0	4	5	8	11	12	3	25	51	0	14	2	2	0	10	3	10	0	1	8	0					
19	2	3	0	6	0	0	0	0	0	0	2	0	5	2	7	9	0	23	6	1	0	3	2	1	9	0	4	0					
20	94	125	13	269	53	5	11	15	5	125	4	252	70	116	395	14	6	1250	109	43	0	214	85	534	6	34	309	26					
21	17	45	2	109	0	0	2	8	4	15	1	159	45	10	176	2	1	109	488	0	7	121	8	182	43	0	94	3					
22	0	4	1	3	3	0	1	0	0	0	0	17	0	10	10	2	0	43	0	92	2	0	4	69	0	0	1						
23	0	15	0	7	1	1	0	0	0	0	1	0	1	4	6	29	0	3	0	7	2	63	0	9	46	0	6	2					
24	8	63	0	44	26	1	9	4	25	181	0	263	39	59	333	10	2	214	121	0	0	795	30	219	14	3	150	6					
25	4	11	0	2	3	1	2	5	0	0	0	31	1	16	89	3	1	85	8	4	9	30	226	130	0	0	3	0					
26	139	381	4	465	103	16	11	21	9	100	5	261	106	204	502	10	9	534	182	69	46	219	130	1597	8	42	478	6					
27	3	7	4	8	0	0	0	0	3	9	0	13	14	3	22	0	0	6	43	0	0	14	0	8	56	0	6	1					
28	0	10	0	33	1	6	0	4	0	0	0	6	2	7	16	1	0	34	0	0	6	3	0	42	0	100	24	0					
29	144	172	0	581	20	1	3	8	29	12	8	237	32	114	443	8	4	309	94	0	2	150	3	478	6	24	1021	0					
30	0	5	1	3	0	1	3	3	1	8	0	20	9	6	40	0	0	26	3	1	2	6	0	6	1	0	0	58					
(b) Co-occurrence																																	
ID ↓→	1	2	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
1	1.00	0.44	0.00	0.83	0.02	0.00	0.04	0.06	0.09	0.00	0.01	0.05	0.02	0.13	0.38	0.01	0.01	0.24	0.04	0.00	0.00	0.02	0.01	0.35	0.01	0.00	0.36	0.00					
2	0.24	1.00	0.00	0.56	0.01	0.01	0.04	0.03	0.09	0.02	0.01	0.07	0.02	0.20	0.35	0.00	0.00	0.17	0.06	0.01	0.02	0.08	0.01	0.51	0.01	0.01	0.23	0.01					
4	0.00	0.04	1.00	0.00	0.04	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.21	0.18	0.07	0.39	0.00	0.00	0.46	0.07	0.04	0.00	0.00	0.14	0.14	0.00	0.00	0.04					
5	0.30	0.37	0.00	1.00	0.03	0.00	0.02	0.03	0.08	0.00	0.01	0.13	0.02	0.18	0.41	0.01	0.01	0.24	0.10	0.00	0.01	0.04	0.00	0.42	0.01	0.03	0.52	0.00					
6	0.04	0.05	0.00	0.14	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.07	0.41	0.04	0.00	0.00	0.24	0.00	0.01	0.00	0.12	0.01	0.46	0.00	0.00	0.09	0.00					
7	0.00	0.22	0.06	0.17	0.06	1.00	0.00	0.00	0.00	0.00	0.00	0.06	0.06	0.11	0.33	0.00	0.00	0.28	0.00	0.00	0.06	0.06	0.06	0.89	0.00	0.33	0.06	0.06					
8	0.27	0.53	0.00	0.47	0.02	0.00	1.00	0.04	0.07	0.09	0.07	0.04	0.05	0.29	0.49	0.00	0.00	0.20	0.04	0.02	0.00	0.16	0.04	0.20	0.00	0.00	0.05	0.05					
10	0.33	0.29	0.00	0.43	0.00	0.00	0.03	1.00	0.07	0.01	0.00	0.20	0.16	0.07	0.61	0.00	0.00	0.21	0.11	0.00	0.00	0.06	0.07	0.30	0.00	0.06	0.11	0.04					
11	0.23	0.46	0.00	0.57	0.00	0.00	0.03	0.03	1.00	0.09	0.01	0.25	0.00	0.13	0.62	0.03	0.00	0.03	0.03	0.00	0.00	0.17	0.00	0.06	0.02	0.00	0.19	0.01					
12	0.00	0.04	0.00	0.00	0.00	0.00	0.01	0.00	0.03	1.00	0.01	0.28	0.16	0.10	0.47	0.01	0.00	0.29	0.03	0.00	0.00	0.42	0.00	0.23	0.02	0.00	0.03	0.02					
13	0.22	0.30	0.00	0.43	0.00	0.00	0.17	0.00	0.09	0.13	1.00	0.04	0.26	0.13	0.57	0.35	0.00	0.17	0.04	0.00	0.00	0.00	0.00	0.22	0.00	0.00	0.35	0.00					
14	0.02	0.05	0.01	0.16	0.02	0.00	0.00	0.02	0.04	0.13	0.00	1.00	0.07	0.04	0.33	0.01	0.01	0.27	0.17	0.02	0.00	0.28	0.03	0.28	0.01	0.01	0.26	0.02					
15	0.03	0.04	0.02	0.06	0.05	0.00	0.01	0.03	0.00	0.22	0.02	0.20	1.00	0.11	0.65	0.04	0.01	0.22	0.14	0.00	0.01	0.12	0.00	0.33	0.04	0.01	0.10	0.03					
16	0.08	0.24	0.00	0.32	0.15	0.00	0.03	0.01	0.03	0.07	0.00	0.06	0.06	1.00	0.40	0.00	0.01	0.19	0.02	0.02	0.01	0.09	0.03	0.33	0.00	0.01	0.18	0.01					
17	0.10	0.17	0.01	0.29	0.01	0.00	0.02	0.03	0.06	0.13	0.01	0.19	0.13	0.16	1.00	0.02	0.01	0.25	0.11	0.01	0.02	0.21	0.06	0.32	0.01	0.01	0.28	0.03					
18	0.04	0.06	0.00	0.12	0.00	0.00	0.00	0.00	0.00	0.10	0.16	0.22	0.24	0.06	0.49	1.00	0.00	0.27	0.04	0.04	0.00	0.20	0.06	0.20	0.00	0.02	0.16	0.00					
19	0.09	0.13	0.00	0.26	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.22	0.09	0.30	0.39	0.00	1.00	0.26	0.04	0.00	0.13	0.09	0.04	0.39	0.00	0.00	0.17	0.00					
20	0.08	0.10	0.01	0.22	0.04	0.00	0.01	0.01	0.00	0.10	0.00	0.20	0.06	0.09	0.32	0.01	0.00	1.00	0.09	0.03	0.00	0.17	0.07	0.43	0.00	0.03	0.25	0.02					
21	0.03	0.09	0.00	0.22	0.00	0.00	0.00	0.02	0.01	0.03	0.00	0.33	0.09	0.02	0.36	0.00	0.00	0.22	1.00	0.00	0.01	0.25	0.02	0.37	0.09	0.00	0.19	0.01					
22	0.00	0.04	0.01	0.03	0.03	0.00	0.01	0.00	0.00	0.00	0.18	0.00	0.11	0.11	0.02	0.00	0.47	0.00	1.00	0.00	0.02	0.00	0.04	0.75	0.00	0.00	0.00	0.01					
23	0.00	0.24	0.00	0.11	0.02	0.02	0.00	0.00	0.00	0.02	0.00	0.02	0.06	0.10	0.46	0.00	0.05	0.00	0.11	0.03	1.00	0.00	0.14	0.73	0.00	0.10	0.03	0.03					
24	0.01	0.08	0.00	0.06	0.03	0.00	0.01	0.01	0.03	0.23	0.00	0.33	0.05	0.07	0.42	0.01	0.00	0.27	0.15	0.00	0.00	1.00	0.04	0.28	0.02	0.00	0.19	0.01					
25	0.02	0.05	0.00	0.01	0.01	0.00	0.01	0.02	0.00	0.00	0.00																						

Table 6: Non-fiction genre misprediction analysis

(a) Sample Count																																
ID ↓→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	27	28	29	30			
1	149	27	38	0	36	4	0	4	0	60	2	0	4	24	14	22	11	2	0	0	4	0	20	0	74	2	0	7	6			
2	27	360	107	2	67	21	6	0	11	12	6	5	2	161	69	36	33	4	6	10	27	39	85	8	46	2	1	26	9			
3	38	107	795	31	27	17	7	29	2	48	14	35	8	378	174	112	159	21	24	123	80	45	21	10	43	22	51	25	147			
4	0	2	31	161	0	5	19	0	0	0	7	0	0	46	84	4	2	0	0	0	14	8	33	0	13	81	0	2	0			
5	36	67	27	0	174	5	0	2	2	19	39	0	7	45	33	13	10	6	5	0	1	0	31	0	66	17	0	50	0			
6	4	21	17	5	5	77	0	0	1	2	1	5	1	29	27	31	6	0	2	3	2	1	18	1	20	2	0	2	1			
7	0	6	7	19	0	0	50	0	0	0	2	0	0	17	20	2	0	1	1	0	0	0	9	0	23	3	2	0	0			
8	4	0	29	0	2	0	0	116	0	16	0	2	44	11	7	8	12	3	1	0	2	0	40	0	8	18	0	2	23			
9	0	11	2	0	2	1	0	0	32	2	0	7	0	7	4	1	1	0	4	0	1	0	26	0	0	5	1	0	0			
10	60	12	48	0	19	2	2	16	2	185	0	0	13	52	36	7	11	2	0	14	8	2	35	0	85	1	9	6	24			
11	2	6	14	7	39	1	0	0	0	0	165	9	24	4	95	9	3	16	9	0	33	0	35	13	9	83	2	14	0			
12	0	5	35	0	0	5	0	2	7	0	9	101	0	28	46	12	25	2	6	11	7	0	18	16	5	16	3	0	2			
13	4	2	8	0	7	1	0	44	0	13	24	0	176	1	66	0	1	46	0	0	21	0	76	1	30	109	5	0	0			
14	24	161	378	46	45	29	17	11	7	52	4	28	1	992	340	58	181	16	24	147	163	59	138	12	156	2	21	43	120			
15	14	69	174	84	33	27	20	7	4	36	95	46	66	340	919	58	159	41	26	42	208	30	179	23	140	210	6	49	26			
16	22	36	112	4	13	31	2	8	1	7	9	12	0	58	58	232	23	0	2	9	4	12	32	1	9	7	10	6	35			
17	11	33	159	2	10	6	0	12	1	11	3	25	1	181	159	23	384	3	18	26	56	18	52	16	9	1	0	17	52			
18	2	4	21	0	6	0	1	3	0	2	16	2	46	16	41	0	3	98	0	6	15	2	24	2	36	39	0	0	0			
19	0	6	24	0	5	2	1	1	4	0	9	6	0	24	26	2	18	0	63	4	6	0	29	0	4	3	0	3	2			
20	0	10	123	0	0	3	0	0	0	14	0	11	0	147	42	9	26	6	4	246	21	20	19	8	14	2	18	0	70			
21	4	27	80	14	1	2	0	2	1	8	33	7	21	163	208	4	56	15	6	21	449	0	68	14	42	125	1	0	14			
22	0	39	45	8	0	1	0	0	0	2	0	0	0	59	30	12	18	2	0	20	0	109	15	1	10	0	1	2	10			
23	20	85	21	33	31	18	9	40	26	35	35	18	76	138	179	32	52	24	29	19	68	15	567	11	65	116	26	28	16			
24	0	8	10	0	0	1	0	0	0	0	13	16	1	12	23	1	16	2	0	8	14	1	11	54	5	16	1	0	1			
25	74	46	43	13	66	20	23	8	0	85	9	5	30	156	140	9	9	36	4	14	42	10	65	5	388	34	0	26	12			
27	2	2	22	81	17	2	3	18	5	1	83	16	109	2	210	7	1	39	3	2	125	0	116	16	34	397	6	4	2			
28	0	1	51	0	0	0	2	0	1	9	2	3	5	21	6	10	0	0	0	18	1	1	26	1	0	6	91	0	17			
29	7	26	25	2	50	2	0	2	0	6	14	0	0	43	49	6	17	0	3	0	0	2	28	0	26	4	0	113	0			
30	6	9	147	0	0	1	0	23	0	24	0	2	0	120	26	35	52	0	2	70	14	10	16	1	12	2	17	0	236			
(b) Co-occurrence																																
ID ↓→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	27	28	29	30			
1	1.00	0.18	0.26	0.00	0.24	0.03	0.00	0.03	0.00	0.40	0.01	0.00	0.03	0.16	0.09	0.15	0.07	0.01	0.00	0.00	0.03	0.00	0.13	0.00	0.50	0.01	0.00	0.05	0.04			
2	0.08	1.00	0.30	0.01	0.19	0.06	0.02	0.00	0.03	0.03	0.02	0.01	0.01	0.45	0.19	0.10	0.09	0.01	0.02	0.03	0.08	0.11	0.24	0.02	0.13	0.01	0.00	0.07	0.03			
3	0.05	0.13	1.00	0.04	0.03	0.02	0.01	0.04	0.00	0.06	0.02	0.04	0.01	0.48	0.22	0.14	0.20	0.03	0.03	0.15	0.10	0.06	0.03	0.01	0.05	0.03	0.06	0.03	0.18			
4	0.00	0.01	0.19	1.00	0.00	0.03	0.12	0.00	0.00	0.00	0.04	0.00	0.00	0.29	0.52	0.02	0.01	0.00	0.00	0.00	0.09	0.05	0.20	0.00	0.08	0.50	0.00	0.01	0.00			
5	0.21	0.39	0.16	0.00	1.00	0.03	0.00	0.01	0.01	0.11	0.22	0.00	0.00	0.04	0.26	0.19	0.07	0.06	0.03	0.03	0.00	0.01	0.00	0.18	0.00	0.38	0.10	0.00	0.29	0.00		
6	0.05	0.27	0.22	0.06	0.06	1.00	0.00	0.00	0.01	0.03	0.01	0.06	0.01	0.38	0.35	0.40	0.04	0.08	0.00	0.00	0.04	0.03	0.01	0.23	0.01	0.26	0.03	0.00	0.01			
7	0.00	0.12	0.14	0.38	0.00	0.00	1.00	0.00	0.00	0.04	0.00	0.00	0.00	0.34	0.40	0.04	0.00	0.02	0.02	0.00	0.00	0.00	0.18	0.00	0.46	0.06	0.04	0.00	0.00			
8	0.03	0.00	0.25	0.00	0.02	0.00	0.00	1.00	0.00	0.14	0.00	0.02	0.38	0.09	0.06	0.07	0.10	0.03	0.01	0.00	0.02	0.00	0.34	0.00	0.07	0.16	0.00	0.02	0.20			
9	0.00	0.34	0.06	0.00	0.06	0.03	0.00	0.00	1.00	0.06	0.00	0.22	0.00	0.22	0.13	0.03	0.03	0.00	0.13	0.00	0.03	0.00	0.81	0.00	0.00	0.16	0.03	0.00	0.00			
10	0.32	0.06	0.26	0.00	0.10	0.01	0.01	0.09	0.01	1.00	0.00	0.00	0.07	0.28	0.19	0.04	0.06	0.01	0.00	0.08	0.04	0.01	0.19	0.00	0.46	0.01	0.05	0.03	0.13			
11	0.01	0.04	0.08	0.04	0.24	0.01	0.00	0.00	0.00	0.00	1.00	0.05	0.15	0.02	0.58	0.05	0.02	0.10	0.05	0.00	0.20	0.00	0.21	0.08	0.05	0.50	0.01	0.08	0.00			
12	0.00	0.05	0.35	0.00	0.00	0.05	0.00	0.02	0.07	0.00	0.09	1.00	0.00	0.28	0.46	0.12	0.25	0.02	0.06	0.11	0.07	0.00	0.18	0.16	0.05	0.16	0.03	0.00	0.02			
13	0.02	0.01	0.05	0.00	0.04	0.01	0.00	0.25	0.00	0.07	0.14	0.00	1.00	0.01	0.38	0.00	0.01	0.26	0.00	0.00	0.12	0.00	0.43	0.01	0.17	0.62	0.03	0.00	0.00			
14	0.02	0.16	0.38	0.05	0.05	0.03	0.02	0.01	0.01	0.05	0.00	0.03	0.00	1.00	0.34	0.06	0.18	0.02	0.02	0.15	0.16	0.06	0.14	0.01	0.16	0.00	0.02	0.04	0.12			
15	0.02	0.08	0.19	0.09	0.04	0.03	0.02	0.01	0.00	0.04	0.10	0.05	0.07	0.37	1.00	0.06	0.17	0.04	0.03	0.05	0.23	0.03	0.19	0.03	0.15	0.23	0.01	0.05	0.03			
16	0.09	0.16	0.48	0.02	0.06	0.13	0.01	0.03	0.00	0.03	0.04	0.05	0.00	0.25	0.25	1.00	0.10	0.00	0.01	0.04	0.02	0.05	0.14	0.00	0.04	0.03	0.04	0.03	0.15			
17	0.03	0.09	0.41	0.01	0.03	0.02	0.00	0.03	0.00	0.03	0.01	0																				

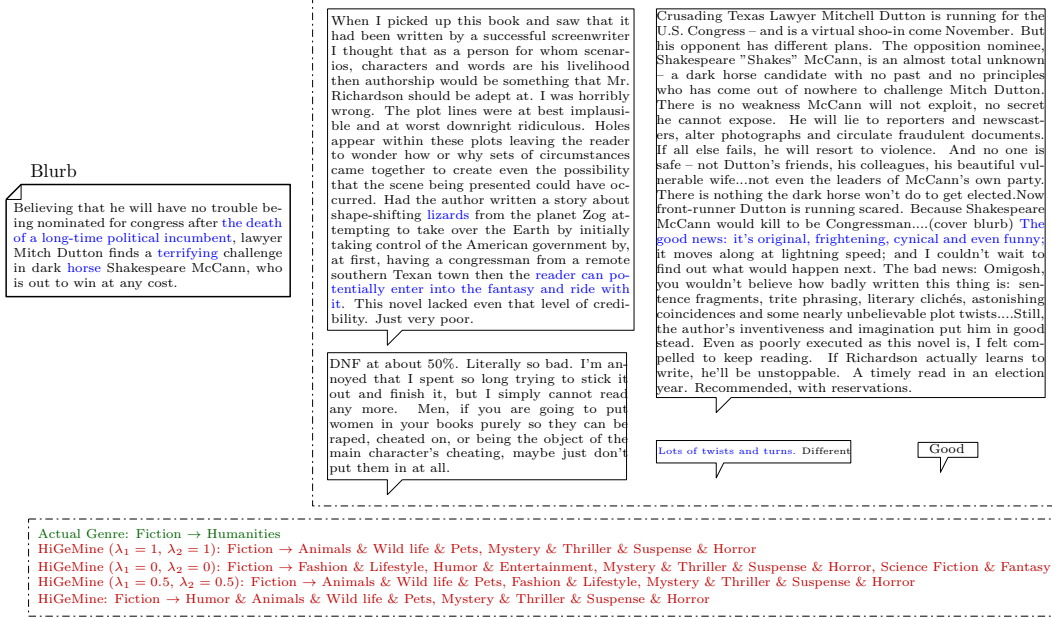


Figure 4: Qualitative analysis for misprediction: misleading genre indicative phrases in blurbs and reviews (highlighted in blue) may trigger systematic prediction inaccuracies across all variation of HeGeMine.

in blurbs and reviews that trigger systematic prediction inaccuracies across all model configurations, HiGeMine (1, 1), HiGeMine (0, 0), HiGeMine (0.5, 0.5), and standard HiGeMine. The segments highlighted blue show how the specific genre-indicative terms in both modalities result in consistent misclassifications despite the varying modality ratios of the model. This underscores the need for improved noise detection mechanisms and context-sensitive feature weighting to enhance the robustness of the model against adversarial textual patterns.

Table 5 (a) and Table 6 (a) present pairwise genre co-occurrence counts in the training set corresponding to *fiction* (F) and *non-fiction* (NF), where cell (ID_i, ID_j) represents samples containing both genre ID_i and ID_j . Table 5 (b) and Table 6 (b) show co-occurrence ratios between genres, quantifying how frequently genre ID_i appears given genre ID_j . Finally, Table 5 (c) and Table 6 (c) report the misprediction rate of genre ID_i misclassified as genre ID_j .

Our analysis from Table 5 and Table 6 highlights that frequent co-occurrence between the two genres, coupled with the significant differences in the number of samples in one genre without association with another, causes a high misprediction rate. For instance, in Table 6 (a) NF: *craft & hobbies & home* (genre ID: 9) and NF: *reference & language* (genre ID: 23) appeared in 26 samples together. However, we have a total of 32 samples associated with genre ID: 9, which leaves only six genre ID: 9 and non-ID: 23 samples. Conversely, we have a total of 567 samples associated with genre ID: 23, leaving 541 samples of genre ID: 23 and non-ID: 9. It is creating an imbalance between genres ID: 9 and non-ID: 23 and non-ID: 9 ID: 23, leading to classify genre ID: 9 to genre ID: 23, with the misclassification rate of

one. The traditional data augmentation methods are limited in their ability to address this imbalance issue, as increasing the number of samples for genre ID: 9 genre also increases the genre ID: 23 samples due to the inter-genre relation. Furthermore, in Table 5, we observe a similar issue on genre ID pairs: (30 & 17), (15 & 17), and (19 & 20).

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